

Requirements — Functional & Non-Functional (multi-OS)

Exigences pour le système multi-OS (AIX · Linux distros · Windows). FR = fonctionnelles, NFR = non fonctionnelles. Voir aussi [requirements.csv](#).

Functional Requirements

Id	Requirement	Priority	Linked Objective	Acceptance Criteria
FR-01	The system MUST ingest one canonical OS-agnostic JSON document (Storage Schema v2) per host, validating it against a single shared schema regardless of <code>platform</code> (aix linux windows).	Must	1,4	Given a v2 document for any OS, ingestion validates required identity fields (<code>host_id</code> , <code>hostname</code> , <code>platform</code> , <code>os_family</code> , <code>os_distribution</code> , <code>os_version</code> , <code>snapshot_id</code> , <code>scan_timestamp</code>); malformed or schema-violating documents are rejected with a typed error and never partially persisted.
FR-02	The system MUST collect storage state by branching the collection command per <code>platform</code> , with AIX, Linux, and Windows implemented now at the same depth and obligation level (no phase 2).	Must	1,3,4	AIX uses <code>lsvg / lslv / lspv / df -k / df -v / lspv -a</code> ; Linux uses <code>df / df -i / vgs / lvs / pvs --units b / tune2fs / xfs_info / btrfs</code> ; Windows uses WinRM+PowerShell (<code>Get-Volume</code> , <code>Get-Partition</code> , <code>Get-Disk</code> , <code>Get-PhysicalDisk</code> , <code>Get-StoragePool</code> , <code>Get-VirtualDisk</code> , <code>Win32_PageFileUsage</code> , <code>Get-Acl</code>). Each branch emits a complete v2 document; a missing Windows path fails CI the same as a missing AIX/Linux path.

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FR-03	The Windows collector MUST connect over WinRM and execute PowerShell to populate <code>filesystems[]</code> , <code>storage_topology</code> , and <code>paging[]</code> as a first-class platform.	Must	1,3,4,5	For a Windows host, NTFS/ReFS volumes (drive letters and folder mount points) appear in <code>filesystems[]</code> with sizes in bytes; Storage Spaces pools map to <code>storage_topology.groups</code> (<code>group_kind=storage-pool</code>) and virtual disks to <code>volumes</code> (<code>volume_kind=virtual-disk</code>); basic disks yield <code>partition</code> volumes with empty groups; pagefile appears in <code>paging[]</code> (<code>kind=windows-pagefile</code>).
FR-04	The system MUST normalize all capacity values to bytes on every platform and field, performing no lossy unit rounding.	Must	1,3,4	All <code>size*_bytes</code> , <code>reserved_bytes</code> , <code>extent_size_bytes</code> , and <code>paging</code> size fields are integers in bytes; AIX <code>df -k / -v</code> kilobytes and any PowerShell values are converted exactly; round-trip checks confirm no precision loss.
FR-05	The system MUST distinguish nullable fields (applicable-but-unknown / not-applicable for the OS or fstype) from absent/malformed fields.	Must	1,4	<code>reserved_bytes</code> is null on xfs/jfs2/ntfs/btrfs and populated on ext; <code>inodes_total</code> / <code>inodes_used</code> are null on Windows; <code>mount_options</code> null on Windows; <code>acl_summary</code> null on Unix; <code>capabilities</code> AIX fields null where not applicable. A null on a nullable field passes validation; a missing required field is rejected.

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FR-06	The system MUST derive <code>storage_class</code> (system vs data) per filesystem from <code>group_ref / backing_ref</code> linkage, generalized across all OS including Windows.	Must	1,5	AIX rootvg-backed and Linux root/system-volume filesystems classify as <code>system</code> ; datavg and non-root as <code>data</code> ; Windows BootDevice / Windows-install volume (typically <code>C:</code>) classifies as <code>system</code> and other drives as <code>data</code> . Classification is reproducible from the document alone without re-collection.
FR-07	The system MUST compute btrfs (SUSE/SLES) true usage in a subvolume- and snapshot-aware manner, not from plain <code>df</code> .	Must	1,3	For btrfs filesystems, used/available bytes are derived from <code>btrfs filesystem usage</code> /qgroups rather than <code>df</code> ; documents flag the source so a SUSE host with subvolumes reports true bytes, and a deliberate <code>df</code> -vs-qgroup discrepancy test selects the qgroup value.
FR-08	The system MUST link filesystems to storage topology via <code>group_ref / backing_ref</code> keys (not nesting) and expose this linkage for derivation.	Must	1,4	Each filesystem with <code>backing_kind</code> in {lvm-lv, virtual-disk, storage-space} resolves its <code>group_ref / backing_ref</code> to an entry in <code>storage_topology</code> ; unresolved references are reported as data-quality defects; plain-partition Linux and basic-disk Windows correctly carry empty topology groups.
FR-09	The system MUST perform layered reference matching of each host/filesystem/group against standards, in priority order, emitting an explicit <code>no_reference</code> when no layer matches.	Must	1,4,5	Matching attempts layers from most specific (e.g. <code>host_role_env</code> + <code>os_distribution</code> + <code>mount_point</code>) to least specific (platform default); the selected layer and its precedence are recorded; when no layer applies, the item is marked <code>no_reference</code> and excluded from deviation scoring rather than defaulted silently.

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FR-10	The system MUST detect deviations versus standards for both filesystems and storage groups.	Must	1,3,5	For each matched item, the engine flags size/threshold/fstype/mount-option/redundancy deviations with severity and the reference used; AIX/Linux/Windows items are all evaluable; results are reproducible for identical inputs.
FR-11	The system MUST detect storage optimization opportunities (useless, obsolete, misplaced data) using non-sensitive <code>content_aggregates</code> .	Should	2,5	The engine flags candidates such as large reclaimable aggregates, stale data, and data placed on <code>system</code> storage that belongs on <code>data</code> , each with a rationale and without accessing file contents.
FR-12	The system MUST flag inappropriate data on system storage for risk/compliance, including Windows <code>C:</code> .	Must	5	Filesystems classified <code>system</code> carrying data-class aggregates are flagged as risk findings with the originating standard; Windows system-volume violations are surfaced identically to AIX rootvg / Linux root.
FR-13	The system MUST produce capacity forecasts and growth trends per filesystem and group across snapshots.	Should	3,4	Given ≥ 2 snapshots per host, the system estimates time-to-full / projected utilization with a confidence indicator; filesystems near saturation are prioritized; forecasts are emitted for all platforms.
FR-14	The system MUST summarize a non-sensitive Windows ACL summary for access compliance, never persisting raw or sensitive ACLs.	Must	5	Windows volumes carry <code>acl_summary</code> (access_model <code>ntfs-acl</code>) capturing non-sensitive access posture; compliance checks evaluate it against standards; raw ACE/SID-level data is not stored or emitted.
FR-15	The system SHOULD use an LLM/agentive (LangGraph) layer to generate explanations, prioritizations, and remediation suggestions grounded in detected findings.	Should	1,2,3,4,5	LLM outputs cite the specific finding, reference layer, and metrics they are based on; no recommendation is produced without an underlying deterministic finding; outputs are advisory only.

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FR-16	The system MUST route material or low-confidence findings and proposed actions through a human-in-the-loop (HITL) approval step.	Must	1,2,5	Findings above a severity/impact threshold or below a confidence threshold require explicit human approval before being marked actionable; approvals/rejections are recorded with actor and timestamp.
FR-17	The system MUST generate fleet- and host-level reports covering deviations, optimization, forecasts, and risk, segmented by OS.	Must	1,2,3,4,5	Reports aggregate findings across >=500 hosts, are filterable by platform/os_distribution/host_role_env/storage_class, and each finding traces to its host, snapshot, and reference layer.
FR-18	The system SHOULD capture feedback on findings and recommendations to improve future matching and detection.	Should	1,4	Users can confirm/dismiss/correct findings; feedback is stored with the finding id and reference layer and is retrievable to tune standards and suppress recurring false positives.
FR-19	The system MUST record <code>collection_method</code> , <code>scan_confidence</code> , and <code>scan_timestamp</code> , and surface low-confidence or stale snapshots.	Must	3,4	Each document carries provenance; snapshots below a confidence threshold or older than a freshness window are flagged in detection and reporting and may be excluded from forecasting.
FR-20	The system MUST normalize <code>os_family / os_distribution</code> so RHEL, Oracle Linux, Rocky, and AlmaLinux share XFS/LVM standards while Debian/Ubuntu (ext4) and SUSE/SLES (btrfs) apply their own.	Must	1,4	A document tagged OracleLinux is matched against the same RHEL-family XFS standard set (no <code>reserved_bytes</code> expected); Debian/Ubuntu match ext4 standards (with <code>reserved_bytes</code>); SUSE/SLES match btrfs subvolume-aware standards.

Non-Functional Requirements

Id	Category	Requirement	Target/Metric
NFR-01	Portability	One schema and pipeline MUST support AIX, RHEL, Oracle Linux, Rocky, AlmaLinux, Debian, Ubuntu, SUSE/SLES (btrfs), and Windows as first-class platforms delivered together.	100% of the 5 objectives evaluable on every listed OS; zero OS gated to a later phase; Windows collection and standards shipped in the same release as AIX/Linux.
NFR-02	Portability	Per-platform collection branches MUST be independently testable with representative fixtures for each OS/fstype.	Golden-fixture coverage for AIX/JFS2, XFS, ext4, btrfs, NTFS, and ReFS; CI fails if any platform branch lacks a passing fixture.
NFR-03	Performance	Ingestion, validation, and deviation detection of a single host document MUST complete within an interactive budget.	<= 2 s p95 per host document end-to-end (ingest + validate + match + detect), excluding LLM steps.
NFR-04	Performance	A full fleet detection pass MUST complete within an operational window.	>= 500 host documents processed in <= 15 min on the reference deployment; linear or better scaling.
NFR-05	Scalability	The system MUST scale to a fleet of at least 500 servers with mixed OS and multi-snapshot history.	Sustained ingestion of 500+ hosts with N snapshots each without architectural change; horizontal scale-out supported.
NFR-06	Security/Privacy	The system MUST process metadata only; no file contents are collected, stored, or transmitted.	Static and runtime checks confirm only non-sensitive aggregates/metadata are persisted; presence of file-content data fails the pipeline.
NFR-07	Security/Privacy	Windows collection MUST persist only a non-sensitive ACL summary, never raw ACLs/ACEs/SIDs.	Stored <code>acl_summary</code> contains no raw ACE/SID detail; automated check rejects any document containing raw ACL data.
NFR-08	Security/Privacy	All remote collection MUST be encrypted, including WinRM for Windows.	WinRM uses HTTPS/encrypted transport; SSH for Unix; plaintext transport rejected; credentials never stored in documents.
NFR-09	Reliability	Collection and ingestion MUST degrade gracefully on partial failure, never producing partially-persisted documents.	Partial collection yields a flagged-incomplete document or controlled failure; atomic persistence (all-or-nothing) per host snapshot.
NFR-10	Maintainability	Standards, reference layers, and OS branches MUST be configurable/extensible without code changes to core logic.	New OS standard or reference layer added via configuration; adding a distro that reuses a family standard requires no engine code change.

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NFR-11	Observability	The pipeline MUST emit metrics and structured logs for collection, ingestion, matching, and detection per platform.	Per-stage success/failure counts, latencies, and <code>no_reference</code> rates exposed; per-platform dashboards available.
NFR-12	Auditability	Every finding and recommendation MUST be traceable to host, snapshot, reference layer, and (if any) HITL decision.	100% of findings carry lineage (host_id, snapshot_id, reference layer, rule id); HITL approvals/rejections logged with actor and timestamp; audit trail immutable.
NFR-13	Data Quality	Capacity data MUST be lossless (bytes), and FS<->topology linkage MUST be verifiable, with full collection lineage.	All sizes integer bytes with no rounding; unresolved <code>group_ref</code> / <code>backing_ref</code> flagged as defects; lineage (<code>collection_method</code> , <code>scan_confidence</code> , <code>scan_timestamp</code>) present on every document.
NFR-14	Data Quality	btrfs sizing MUST be reconciled against qgroups so reported bytes are true, not <code>df</code> - misleading.	For btrfs hosts, reported used/available match qgroup-derived values within tolerance; <code>df</code> -only sizing rejected for btrfs.
NFR-15	Cost	The pipeline MUST bound LLM usage to control cost without degrading deterministic detection.	Deterministic detection runs with zero LLM cost; LLM invoked only for explanation/prioritization on filtered findings; per-run token/cost budget enforced.
NFR-16	Usability	Reports and HITL interfaces MUST let operators filter, drill down, and act across the mixed fleet.	Operators filter by platform/os_distribution/host_role_env/storage_class and drill from fleet to host to filesystem to reference in <= 3 interactions.
NFR-17	Reliability	Snapshot freshness and confidence MUST gate downstream use.	Stale (beyond freshness window) or low- <code>scan_confidence</code> snapshots are flagged and optionally excluded from forecasting/scoring per policy.